Benchmark Coverage Gap: A Systematic Analysis of Real-World AI Capabilities and Evaluation Practices

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1 Abstract

Large language models (LLMs) are increasingly deployed across professional, educational, and personal contexts, with evaluation practice centred on standardised benchmarks such as the Massive Multitask Language Understanding (MMLU) benchmark, HumanEval, and GSM8K. A substantial and growing body of evidence suggests, however, that strong benchmark performance does not reliably predict utility on the tasks that users actually bring to these systems. This project addresses the absence of a systematic, usage-weighted analysis of benchmark coverage across the full range of documented real-world LLM capabilities.

Using a mixed-methods design comprising inductive thematic analysis of empirical usage data, systematic benchmark inventory construction, quantitative gap score analysis, and academic literature validation, the project produces four principal outputs. First, an eight-capability taxonomy — covering Content Generation, Code Development and Technical Problem Solving, Information Retrieval and Advisory, Learning and Education Support, Review and Feedback, Translation and Language Processing, Data Analysis and Summarisation, and Conversational Interaction and Roleplay — validated with a Cohen's of 1.00 against the Anthropic Economic Index top-task data. Second, a structured inventory of 28 major LLM benchmarks with complete metadata and quality ratings across five dimensions. Third, a usage-weighted coverage analysis demonstrating through chi-square testing (^{2} = 31.96, p ¡ 0.001) that benchmark coverage is not distributed in proportion to usage demand, with Code Development (gap score 0.2307), Content Generation (0.2202), and Information Retrieval and Advisory (0.1453) identified as the three most urgent gaps. Fourth, five benchmark design specifications — MaintBench, WorkWriteBench, GroundedAdviceBench, TutorScaffoldBench, and MessyDataBench — and a reusable practitioner Assessment Toolkit. The findings provide the research community with an evidence-based diagnostic of where new evaluation investment is most urgently needed.

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3 Chapter 1: Introduction and Background

3.1 1.1 Background and Motivation

Large language models (LLMs) have moved rapidly from research prototypes to deployed tools used by millions of people across professional, educational, and personal contexts. This transition has generated a pressing practical question: how well do the evaluation frameworks developed by the research community actually reflect what users ask these models to do?

The dominant answer, reflected in press releases, model documentation, and public leaderboards, has been to cite performance on standardised benchmarks. Models are routinely reported to achieve high scores on the Massive Multitask Language Understanding (MMLU) benchmark (Hendrycks et al., 2021), on competitive programming tasks such as HumanEval and LiveCodeBench (Jain et al., 2024), and on mathematical reasoning datasets such as GSM8K (Cobbe et al., 2021). These scores are widely used to compare models, to justify deployment decisions, and to direct research investment toward particular capability areas. Yet a growing body of evidence suggests that strong benchmark performance does not reliably translate to strong performance on the tasks that users actually bring to these systems.

Handa et al. (2025) analysed approximately four million Claude.ai conversations mapped to occupational task categories drawn from the O*NET Detailed Work Activities database. Their analysis revealed that technical assistance accounts for approximately 65.1% of real-world usage, reviewing and improving existing work accounts for 58.9%, information retrieval accounts for 16.6%, and summarisation accounts for 16.6%. Critically, reviewing work — the second most common pattern of use — has no dedicated evaluation framework. Users regularly ask models to edit drafts, provide structured feedback on code or writing, and improve the quality of existing content, yet the benchmark ecosystem provides almost no standardised means of measuring how well models perform these tasks.

This disconnect has practical consequences. Xu et al. (2025) demonstrate in the TheAgentCompany benchmark that state-of-the-art models, despite strong performance on standard evaluations, complete only approximately 30% of autonomous workplace tasks. Singh et al. (2025) document systematic selective disclosure of benchmark results, showing that published leaderboard rankings are distorted by up to 112 positions when all results rather than selected subsets are considered. Pezeshkpour and Hruschka (2024) show that simply changing the order of answer options in multiple-choice questions shifts model rankings by approximately eight positions, revealing sensitivity to format rather than genuine capability. Balloccu et al. (2024) document widespread data contamination in closed-source models, and Jain et al. (2024) show that model performance on programming problems drops sharply after training cutoff dates, providing evidence that strong scores often reflect memorisation rather than generalisation.

Together, these findings motivate a systematic investigation into the gap between what benchmarks measure and what users need. Despite substantial individual evidence of misalignment, no existing study provides a comprehensive, usage-weighted map of the entire benchmark ecosystem that quantifies where coverage gaps are most severe and most consequential. This thesis addresses that gap.

3.2 1.2 Problem Statement

The central problem addressed in this thesis is the systematic misalignment between LLM evaluation practice and empirically documented patterns of real-world use. Standard benchmarks measure a narrow and historically determined set of capabilities — predominantly multiple-choice knowledge recall, algorithmic coding, and formal mathematical reasoning — while real users employ LLMs for a broader range of tasks that resist easy operationalisation, such as reviewing and improving content, providing grounded advisory responses, and supporting learning across diverse domains.

This misalignment is not merely an academic concern. When evaluation frameworks do not reflect use, research investment flows toward capabilities that score well on benchmarks rather than capabilities that matter most in deployment. Model selection decisions based on benchmark rankings may not identify the best model for a given practical purpose. Failure modes in deployed systems go undetected until they manifest as real-world errors with real-world consequences.

The specific gap in the literature that this project addresses is the absence of a systematic, empirically grounded, usage-weighted analysis of benchmark coverage across the full range of documented real-world LLM capabilities. Existing critiques of individual benchmarks are numerous and well-founded, but they do not provide the kind of comprehensive diagnostic that practitioners and researchers would need to understand where the evaluation ecosystem as a whole is most deficient and what new benchmarks should be prioritised.

3.3 1.3 Research Objectives

This project pursues nine specific objectives designed to produce a complete empirical account of the benchmark coverage gap and actionable recommendations for closing it.

O1: Build an empirically grounded capability taxonomy derived from real-world LLM usage data, covering a minimum of 95% of documented usage patterns and validated through inter-coder reliability analysis (target Cohen's ¿ 0.8).

O2: Compile a standardised inventory of 15–20 major LLM benchmarks spanning 4–5 capability areas, with complete structured metadata for each benchmark including task format, evaluation metrics, dataset characteristics, quality ratings across five dimensions, and contamination risk assessments.

O3: Map all benchmarks to all capabilities in the taxonomy through a structured coverage matrix, with justified quality ratings for each benchmark-capability pairing.

O4: Quantify benchmark coverage gaps using a usage-weighted gap score formula that accounts for both the frequency with which each capability is used in practice and the quality and quantity of existing evaluation coverage.

O5: Conduct qualitative deep-dive analysis for a selection of capabilities spanning the coverage spectrum, examining the evaluation landscape, the technical challenges to measurement, the real-world consequences of inadequate evaluation, and requirements for adequate coverage.

O6: Validate the framework and findings against published academic literature and cross-platform usage studies, assessing the robustness and generalisability of the capability taxonomy and gap findings.

O7: Produce three to five detailed benchmark design specifications targeting the highest-priority identified gaps, each including worked example items, dataset composition plans, evaluation methodology, validation strategies, and implementation requirements.

O8: Build a reusable Assessment Toolkit comprising an Excel workbook and PDF documentation enabling practitioners to evaluate benchmark relevance for their specific use case.

O9: Write and submit a complete thesis reporting all phases of this research.

3.4 1.4 Contributions

This thesis makes four contributions to the literature on LLM evaluation.

First, it provides the first comprehensive, empirically grounded capability taxonomy for real-world LLM use derive comprehensive, empirically grounded capability taxonomy for real-world LLM use,The taxonomy identifies eight core capabilities — Content Generation (C01), Code Development and Technical Problem Solving (C02), Information Retrieval and Advisory (C03), Learning and Education Support (C04), Review and Feedback (C05), Translation and

Language Processing (C06), Data Analysis and Summarisation (C07), and Conversational Interaction and Roleplay (C08) — that together cover 100% of the top tasks identified in the Anthropic Economic Index (Handa et al., 2025). The taxonomy is validated with a Cohen's of 1.00 and provides formal definitions, decision rules, and worked examples that enable consistent application in future research.

Second, it provides a systematically constructed benchmark inventory of 28 major LLM benchmarks with complete structured metadata and justified quality ratings across five dimensions: coherence, accuracy, clarity, relevance, and efficiency. The inventory is the first to map benchmarks explicitly to a usage-derived capability taxonomy rather than to researcher-defined categories, and it documents contamination risk, update frequency, and real-world applicability for each entry.

Third, it provides the first usage-weighted quantitative analysis of benchmark coverage gaps across all documented LLM use capabilities. The analysis demonstrates through chi-square testing that benchmark coverage is not distributed in proportion to usage frequency (^{2} = 31.96, p ¡ 0.001), and identifies Code Development and Technical Problem Solving (gap score 0.2307), Content Generation (0.2202), and Information Retrieval and Advisory (0.1453) as the three most urgent gaps.

Fourth, it provides actionable design specifications for five new benchmarks — MaintBench, WorkWriteBench, GroundedAdviceBench, TutorScaffoldBench, and MessyDataBench — and a reusable Assessment Toolkit that practitioners can apply to evaluate benchmark portfolios for their specific deployment contexts.

3.5 1.5 Thesis Structure

The remainder of this thesis is organised as follows. Chapter 2 reviews the background literature on LLM evaluation, covering the development and critique of major benchmarks, empirical studies of real-world LLM use, and existing taxonomies of model capabilities. Chapter 3 describes the research design and methodology, covering the five sequential research phases, the justification for methodological choices, and the ethical framework governing the project. Chapter 4 presents the implementation and results for the three core analytical phases: capability taxonomy development, systematic benchmark inventory, and coverage matrix and gap score analysis. Chapter 6 presents the recommendations outputs: five benchmark design specifications and the Assessment Toolkit. References are consolidated at the end of the document.

4 Chapter 2: Background and Literature Review

4.1 2.1 The Rise of LLM Benchmarking

The evaluation of natural language processing (NLP) systems has a history considerably older than the systems it now primarily assesses. Early NLP evaluation relied on narrow, well-specified tasks — part-of-speech tagging, syntactic parsing, and information extraction — where ground truth was largely unambiguous and performance could be measured against annotated corpora with precision. The shift toward neural language models over the first two decades of the twenty-first century gradually eroded the sufficiency of these task-specific metrics. As models became capable of producing coherent text across many domains, researchers needed comparative frameworks that could characterise model capability in aggregate rather than in isolatioto characterise model capabilities uage Understanding Evaluation (GLUE) benchmark (Wang et al., 2018) and its successor SuperGLUE (Wang et al., 2019) marked a turning point. These benchmarks collected multiple existing tasks — natural language inference, coreference resolution, question answering, and sentiment analysis — into unified evaluation suites and provided a single composite score. The limitation was equally immediate: models trained specifically to perform well on GLUE quickly saturated the benchmark, with systems exceeding human performance on SuperGLUE within approximately three years of its publication (He et al., 2021). Saturation was not evidence of genuine parity with human language ability; it was evidence of benchmark-specific optimisation.

The release of GPT-3 in 2020 (Brown et al., 2020) changed the terms of the evaluation challenge. GPT-3 and the large-scale models that followed demonstrated strong performance across an open-ended range of tasks without task-specific fine-tuning, which meant that evaluation frameworks designed around narrow tasks could no longer serve as a complete account of what models could do. The Massive Multitask Language Understanding (MMLU) benchmark, introduced by Hendrycks et al. (2021), responded to this by collecting 57 academic subject areas and framing evaluation as multiple-choice question answering over this broad knowledge domain. MMLU became rapidly adopted as a reporting standard in model technical reports and academic papers, and it remains one of the most widely cited benchmark scores in LLM evaluation as of 2025 (Singh et al., 2025).

The BIG-Bench effort (Srivastava et al., 2022) extended the multi-task principle further, assembling more than 200 tasks across capabilities including arithmetic, commonsense reasoning, translation, coding, and novel problem types. BIG-Bench Hard (Suzgun et al., 2022) subsequently filtered these to the subset most challenging for contemporary models. Alongside these general-purpose benchmarks, domain-specific evaluation frameworks proliferated. HumanEval (Chen et al., 2021) provided a standardised coding benchmark through 164 Python function completion problems. GSM8K (Cobbe et al., 2021) established a standard for elementary mathematical word problem solving. The HellaSwag (Zellers et al., 2019), ARC (Clark et al., 2018), and TruthfulQA (Lin et al., 2022) benchmarks addressed commonsense inference, scientific knowledge, and truthfulness respectively.

By the mid-2020s, public leaderboards had become central to how models were compared and research priorities set. The Open LLM Leaderboard, Chatbot Arena, and the leaderboards maintained by individual model providers presented ranking tables based on aggregated benchmark scores, and these rankings influenced which models received commercial adoption, which capabilities received research attention, and which papers received acceptance at major venues. The practical consequence of this centralisation was substantial: the benchmark ecosystem had become a de facto standard for communicating model capability, making the validity of that ecosystem a matter of practical significance.

4.2 2.2 Empirical Studies of Real-World LLM Use

Despite the dominance of benchmark-based evaluation, empirical study of how people actually use LLMs was, until recently, sparse. Most early work on user behaviour relied on small survey instruments or limited observational samples that could not support generalisable quantitative claims about the distribution of use cases. This changed as large-scale usage data became available from major providers.

The most significant study for the present project is that of Handa et al. (2025), who analysed approximately four million conversations between users and Claude.ai, Anthropic's commercial assistant product, mapping each conversation to task categories drawn from the Occupational Information Network (O*NET) Detailed Work Activities database. Their analysis revealed that technical assistance accounts for approximately 65.1% of recorded usage, reviewing and improving work accounts for approximately 58.9%, and information retrieval and summarisation each account for approximately 16.6%. The critical finding is that reviewing work — the second most common pattern of use — has no dedicated evaluation framework. The Anthropic Economic Index (AEI) February 2026 update extends this analysis and provides the primary usage frequency data used in the present project.

A complementary dataset is provided by Ouyang et al. (2025), who analyse usage of OpenAI's assistant platform through an NBER Working Paper examining millions of API requests. Their findings confirm several patterns from the Handa et al. analysis: writing, coding, analysis, and tutoring emerge as dominant categories, with individual task frequencies broadly consistent with those from the Anthropic data. This cross-platform consistency suggests that the usage patterns identified in the Anthropic data reflect general patterns of user demand rather than idiosyncrasies of a particular model or interface.

The OpenRouter 100T Token Study contributes a third perspective drawn from a multi-model platform that includes substantial open-source model traffic. This source reveals that roleplay and interactive fiction account for a disproportionately large share of token consumption in open-source contexts — a pattern less visible in enterprise-oriented Anthropic data. Together, these three sources provide a more complete picture of the range of tasks that users bring to LLM systems across different platforms and user populations.

Xu et al. (2025) offer a different form of empirical evidence: a benchmark specifically designed to evaluate LLM performance on realistic workplace tasks rather than on academic exercise formats. TheAgentCompany simulates a software company environment and requires LLM agents to complete tasks drawn from human resources, finance, software engineering, and administrative workflows. The finding that state-of-the-art models complete approximately 30% of these tasks, despite strong scores on conventional benchmarks, constitutes direct evidence of the gap between benchmark performance and practical capability.

4.3 2.3 Benchmark Validity and Known Limitations

Alongside the empirical evidence of usage patterns, a substantial body of work has documented specific validity problems in the benchmark ecosystem. These problems concern not only which capabilities are evaluated but whether the evaluations that exist are measuring what they claim to measure.

Data contamination is among the most extensively documented validity threats. Balloccu et al. (2024) conduct a systematic examination of data contamination in closed-source models, showing that benchmark test items appear in model training data at rates that inflate reported performance. Jain et al. (2024) provide a particularly compelling demonstration of contamination effects in the coding domain. LiveCodeBench collects programming problems organised by date of public release and shows that performance drops substantially on problems released after likely training cutoff dates, even for tasks of nominally similar difficulty. This temporal performance gap provides direct evidence that strong scores on static coding benchmarks reflect, at least in part, familiarity with specific problems rather than general programming ability.

Format sensitivity introduces a further validity problem. Pezeshkpour and Hruschka (2024) demonstrate that changing the order of answer options in multiple-choice questions can shift model rankings by approximately eight positions on MMLU-style evaluations. Since the content and the set of possible answers remain unchanged, a model with genuine knowledge should respond identically regardless of option ordering. The observed ranking instability reveals sensitivity to surface-level presentation features rather than the underlying knowledge the benchmark is intended to measure.

Selective disclosure introduces a further distortion between raw model performance and reported rankings. Singh et al. (2025) examine the systematic practices by which benchmark results are reported, showing that companies selectively cite the benchmarks on which their models perform best. Their analysis of the Artificial Analysis Intelligence Leaderboard shows that including all available benchmark results rather than provider-selected subsets can shift rankings by up to 112 positions.

The concept of Goodhart's Law applies with particular force to LLM benchmarking. As Strathern (1997) elaborates, when a measure becomes a target it ceases to be a good measure: optimising directly for a metric tends to decouple it from the underlying construct it was intended to proxy. Chang et al. (2023) review these dynamics comprehensively, noting that the field faces a structural tendency to over-optimise for measurable proxies at the expense of genuine capability development. Benchmark saturation — the progressive erosion of discriminability as models approach ceiling performance — compounds these problems. MMLU, HumanEval, and many other benchmarks have seen performance levels approach ceiling for leading models as of 2024 and 2025, prompting the introduction of harder variants including MMLU-Pro (Wang et al., 2024), HumanEval+ (Liu et al., 2023), GPQA Diamond (Rein et al., 2023), and Humanity's Last Exam (Phan et al., 2025).

4.4 2.4 Existing Capability Taxonomies

Prior to this project, several attempts had been made to organise the landscape of LLM capabilities into structured taxonomies. These efforts varied in their empirical grounding, intended application, and granularity.

The O*NET Detailed Work Activities system, maintained by the United States Department of Labor, provides a comprehensive taxonomy of work-relevant task categories spanning the full range of occupational activities. O*NET is well-suited to mapping LLM usage to economic and labour-market frameworks, as in the Handa et al. (2025) study. However, the O*NET system was not designed for evaluating AI capabilities, and many of its categories reflect occupational role distinctions that do not map cleanly onto the interaction modes relevant for LLM evaluation.

Ouyang et al. (2025) present a usage-derived taxonomy in their analysis of OpenAI platform traffic, identifying writing, coding, analysis, and tutoring as major categories. This scheme is broadly aligned with the capabilities developed in the present project but is not formally defined with decision rules, its inter-coder reliability is not reported, and it is not systematically linked to any benchmark inventory.

Review papers such as Chang et al. (2023) and Liang et al. (2022) organise capabilities into broad clusters — language understanding, knowledge, reasoning, and generation — but these clusters are researcher-defined and reflect the structure of the existing benchmark landscape rather than empirically documented patterns of user demand. The consequence is that the majority of evaluation taxonomies in the literature are shaped by what is easy to benchmark rather than by what users actually do.

Several evaluation frameworks explicitly focus on agentic capabilities. GAIA (Mialon et al., 2023) and the AgentBench family address tool use and multi-step planning, and TheAgentCompany (Xu et al., 2025) extends this to realistic workplace settings. These frameworks capture an increasingly important dimension of LLM behaviour. The present project treats agentic operation as a modality of task execution rather than as a separate capability category, on the grounds that agents may be executing software development, information retrieval, or data analysis tasks while using tools and multi-step planning.

4.5 2.5 Gaps in the Literature

The preceding review reveals a consistent gap: no prior study provides a systematic, usage-weighted, cross-benchmark coverage analysis that links empirically documented usage frequencies to structured quality assessments of whether existing benchmarks actually measure the capabilities users most demand.

Individual critiques of specific benchmarks are numerous and well-documented, but each addresses a single dimension of a single type of benchmark problem. No work integrates these concerns into a framework that prioritises them according to usage frequency, maps the full range of documented capabilities against the full population of major benchmarks, or produces actionable gap scores to direct new benchmark development.

Empirical usage studies have documented what users do with LLMs, but they do not analyse the evaluation implications of their findings. They do not identify which usage capabilities have no corresponding benchmark framework, nor do they quantify the severity of coverage shortfalls relative to demand. Benchmark surveys and evaluation reviews document the landscape of available benchmarks and identify desirable properties for evaluation, but they do not ground their analyses in empirical usage data, and their capability categorisation schemes reflect the existing benchmark structure rather than challenging it.

This project addresses the gap between these bodies of work. By building a capability taxonomy from usage data before examining the benchmark landscape, it ensures that the categories used to assess coverage are driven by user demand rather than by evaluation convenience. By computing usage-weighted gap scores, it produces a prioritised assessment of where new evaluation frameworks are most urgently needed. And by producing actionable benchmark design specifications and a reusable practitioner toolkit, it provides outputs that can directly inform the next generation of LLM evaluation development.

5 Chapter 3: Research Design and Methodology

5.1 3.1 Overall Research Design

This project adopts a mixed-methods research design that combines systematic literature review, qualitative thematic analysis, quantitative gap assessment, and academic literature validation. The rationale for this combination is grounded in the nature of the research problem. Determining whether benchmark coverage is misaligned with real-world use requires, first, an empirically derived account of what users actually do with LLMs, which is an inherently interpretive task suited to qualitative methods; and second, a systematic, measurable characterisation of how well existing benchmarks address those uses, which requires quantitative comparison. Neither method alone is sufficient.

Two alternative designs were considered and rejected. A pure expert survey design — in which practitioners were asked directly to identify evaluation gaps — would depend on the opinions of a small convenience sample rather than on empirical usage data, introducing social desirability bias and limiting the generalisability of findings. A purely computational approach — for example, automated text-matching between benchmark task descriptions and usage logs — would require a pre-existing taxonomy to define the matching categories and would not capture the qualitative judgements about coverage quality that distinguish a benchmark that superficially covers a capability from one that measures it well. The mixed-methods design addresses both limitations.

Validation of the framework and findings was conducted through triangulation against published academic literature and cross-platform usage studies rather than through an expert survey. This decision was made on methodological grounds: the published literature provides a more substantial and generalisable evidence base for assessing the robustness of the capability taxonomy and gap rankings than a small convenience sample of survey respondents. The cross-platform coverage of the Anthropic, OpenAI, and OpenRouter usage datasets ensures that the taxonomy and gap findings are not idiosyncratic to a single provider or user population. The detailed examination of benchmark validity problems in the academic literature provides the evidentiary basis for the quality ratings and coverage assessments that underpin the gap analysis.

5.2 3.2 Five-Phase Research Structure

The project is structured as five sequential phases. Each phase produces the inputs that the next phase requires, so no phase could begin before its predecessor was complete. This design decision was deliberate: beginning the benchmark inventory before the taxonomy was finalised would have required remapping benchmarks to revised capability definitions after the fact, introducing inconsistency. Similarly, running the coverage analysis before the inventory was complete would have produced gap scores based on a partial dataset.

Phase 1 (Weeks 1–4) comprised capability framework development. The goal was to derive an empirically grounded taxonomy of real-world LLM capabilities through systematic thematic analysis of usage data. The outputs of this phase — the eight-capability taxonomy and its decision rules — form the analytic categories used in every subsequent phase.

Phase 2 (Weeks 5–9) comprised the benchmark inventory. This phase compiled a standardised inventory of 28 major LLM benchmarks, each mapped to the Phase 1 taxonomy and assessed against a five-dimension quality rubric.

Phase 3 (Weeks 10–13) comprised the coverage analysis. This phase constructed the capability-benchmark coverage matrix, computed usage-weighted gap scores for all eight capabilities, conducted three statistical analyses, produced five visualisations, and documented qualitative deep-dive analyses and real-world case studies.

Phase 4 (Weeks 14–16) was designated for validation. Rather than deploying an expert survey, this phase conducted structured academic validation: assessing the robustness of the taxonomy and gap findings through triangulation with published usage studies (Handa et al., 2025; Ouyang et al., 2025) and examining the extent to which the literature independently corroborates the coverage gap priorities identified in Phase 3.

Phase 5 (Weeks 17–20) produced the recommendations and toolkit outputs, translating the validated Phase 3 findings into five benchmark design specifications and a practitioner Assessment Toolkit.

5.3 3.3 Qualitative Methods: Thematic Analysis

5.3.1 3.3.1 Rationale for Thematic Analysis

The primary qualitative method used in Phase 1 was the six-phase thematic analysis framework developed by Braun and Clarke (2006). Thematic analysis was chosen over two principal alternatives. Grounded theory, as described by Glaser and Strauss (1967), shares the goal of deriving categories inductively from data but imposes a more prescriptive set of procedures — theoretical sampling, constant comparison, and saturation — that are better suited to iterative interview data than to a pre-assembled corpus of usage logs and occupational task records. Content analysis, as used in quantitative text research (Krippendorff, 2018), requires a pre-existing coding scheme and is primarily concerned with counting occurrences of predefined categories rather than identifying latent themes. Braun and Clarke's framework was chosen because it is transparent and systematic enough to ensure replicability, flexible enough to accommodate a mixed corpus of empirical sources, and well-suited to deriving themes from patterns of meaning rather than surface text features.

5.3.2 3.3.2 Data Corpus

The thematic analysis was conducted on a corpus of 131 task instances drawn from three sources. The Anthropic Economic Index (AEI) February 2026 update (Handa et al., 2025) contributed 103 task instances derived from the mapping of approximately four million Claude.ai conversations to O*NET Detailed Work Activities. This source was treated as the primary empirical anchor because it is the largest published usage dataset for a major commercial LLM and provides quantitative usage frequency estimates for each task type. Ouyang et al. (2025) contributed 19 task category rows from an analysis of OpenAI platform usage patterns, providing cross-platform validation. The OpenRouter 100T Token Study contributed 9 category rows from a cross-model platform, capturing usage patterns beyond single-provider data. The three sources together covered occupational, personal, and educational contexts across multiple major LLM providers.

The WILDCHAT dataset (Zhao et al., 2024), a publicly available corpus of anonymised ChatGPT conversations, was considered for inclusion but excluded on methodological grounds. WILDCHAT conversations are not pre-categorised into task types, and the sampling and manual coding required to extract comparable task instances would have introduced a different layer of analyst interpretation at the data collection stage rather than at the coding stage.

5.3.3 3.3.3 The Six-Phase Procedure

The analysis followed the six phases prescribed by Braun and Clarke (2006) in strict sequence. The familiarisation phase involved reading all 131 task instances in full and recording initial impressions in a memo of approximately 500 words. The memo noted that technical and software tasks dominated the Anthropic data, that professional writing and business communication formed a visible secondary cluster, and that creative and conversational interaction tasks were visible in the OpenRouter data but absent from the occupationally framed AEI categories.

Open coding assigned a short descriptive label to each task instance in the researcher's own words, without theoretical interpretation. Labels included expressions such as ”fix syntax errors in Python script”, ”draft a professional email to a client”, ”explain photosynthesis for a student”, and ”translate a legal document from French to English”. Axial coding grouped the 131 open codes into 19 intermediate categories, revealing several convergences that were non-obvious from surface labels. Software engineering, web development, debugging, DevOps, and agentic technical execution all shared the defining characteristic that the expected output was an executable artefact or technical configuration.

Selective coding collapsed the 19 axial categories into eight core capabilities using the criterion that two axial categories should remain distinct if they would require different evaluation approaches — that is, if they would be assessed by different benchmarks using different metrics. This criterion was applied most carefully in two cases. Content Generation (C01) and Review and Feedback (C05) were kept separate because the model's role differs categorically: as author in C01 and as reviewer or editor in C05. Information Retrieval and Advisory (C03) and Learning and Education Support (C04) were kept separate because the user's primary goal differs: obtaining a complete answer in C03 versus acquiring understanding through guided engagement in C04. Formal definitions, decision rules, and worked examples were produced for each of the eight capabilities, and the final taxonomy document was compiled.

5.3.4 3.3.4 Inter-Coder Reliability

To assess the reliability of the classification scheme, a 10% subsample of the coded data was independently re-coded by a second coder using only the task description and the if/then decision rules from the capability definitions document, without access to the primary coder's labels. Cohen's was calculated using the formula = (P_o P_e) / (1 P_e), where P_o is the observed agreement proportion and P_e is the agreement expected by chance. The result was = 1.0000, indicating perfect agreement and substantially exceeding the project target of ¿ 0.80 (Cohen, 1960). This result demonstrates that the decision rules are sufficiently precise to support consistent application by different coders.

5.4 3.4 Quantitative Methods

5.4.1 3.4.1 Benchmark Quality Rating Rubric

Each benchmark in the Phase 2 inventory was assessed against a five-dimension quality rubric on a 1–5 scale. The five dimensions were: Coherence (the internal consistency and logical structure of the task set); Accuracy (the reliability of the ground truth labels and expected outputs); Clarity (the precision and unambiguity of task instructions); Relevance (the correspondence between what the benchmark measures and real-world capability as defined by the Phase 1 taxonomy); and Efficiency (the practical cost and complexity of running the evaluation). A score of 5 indicated that the benchmark fully met the standard for that dimension; 4 indicated minor issues; 3 indicated notable but manageable issues; 2 indicated significant problems; and 1 indicated a fundamental flaw. Every non-trivial rating was required to be accompanied by a written justification, ensuring that ratings reflected documented evidence rather than unconstrained researcher judgement.

5.4.2 3.4.2 Gap Score Formula

The central quantitative measure of this project is the usage-weighted gap score, defined as:

Gap Score = Usage_Frequency × (1 Normalised_Coverage_Score)

where Usage_Frequency is the proportion of total LLM usage attributable to the capability (derived from summed AEI task data), and Normalised_Coverage_Score is the total coverage score divided by the maximum possible total coverage score, scaled to the range 0–1. A higher gap score indicates a capability that is both heavily used in practice and poorly covered by existing benchmarks, and therefore represents a higher priority for new evaluation development.

5.4.3 3.4.3 Statistical Analyses

Three statistical analyses were conducted. A Pearson correlation was computed between usage frequency and total coverage score across the eight capabilities, testing the null hypothesis that usage frequency and benchmark coverage are uncorrelated. A chi-square goodness-of-fit test assessed whether the observed distribution of benchmark counts across capabilities matched the expected distribution if benchmarks were allocated in proportion to usage frequency. Separate linear regressions were computed for each capability, modelling the annual count of benchmarks published from 2020 to 2025 as a function of year, with the aim of identifying whether benchmark activity in each capability area was increasing, stable, or declining. All statistical operations were conducted in Python using the scipy and numpy libraries, with np.random.seed(42) set for all operations involving randomisation, ensuring exact replicability.

5.5 3.5 Data Sources

Three primary empirical sources were used for the capability taxonomy and usage frequency estimates. The Anthropic Economic Index (AEI) February 2026 update (Handa et al., 2025) provided the primary usage frequency data, mapping approximately four million Claude.ai conversations to occupational task categories and reporting usage percentages for the top 103 tasks. This source was chosen as the primary anchor because it is the largest published usage log for a commercial LLM with task-level quantification. The Ouyang et al. (2025) NBER Working Paper 34255 provided 19 task category rows from an analysis of OpenAI assistant usage, used for cross-platform validation of the taxonomy. The OpenRouter 100T Token Study provided nine category rows from aggregate usage data across a multi-model platform, capturing roleplay and open-source usage patterns underrepresented in the Anthropic data.

For the benchmark inventory, data were drawn from four search channels: Papers with Code benchmark listings, major model technical reports (GPT-4, Claude 3/3.5, Gemini, Llama 2/3, and DeepSeek R1), Google Scholar searches for terms including "LLM benchmark" and "language model evaluation" filtered to 2020–2025, and snowball sampling from the reference lists of Handa et al. (2025), Xu et al. (2025), and Singh et al. (2025).

5.6 3.6 Bias Controls and Replicability

Several procedural controls were applied to limit the effects of researcher bias and ensure replicability. All capability classification decisions in Phase 1 were governed by explicit if/then rules recorded before coding of the main dataset began, preventing post-hoc rationalisation of individual coding decisions. Every quality rating in the Phase 2 benchmark database and every non-zero cell in the Phase 3 coverage matrix was required to be accompanied by a written justification, limiting the influence of unconstrained intuition on the quantitative measures. All statistical operations involving randomisation used np.random.seed(42), ensuring exact reproduction of numerical results. All data files, scripts, and output documents were tracked under Git version control with descriptive commit messages, and all data files and analysis scripts are to be released publicly on GitHub following thesis submission, enabling independent replication of all quantitative findings.

5.7 3.7 Ethical Considerations

The primary empirical data sources — the Anthropic AEI dataset, the Ouyang et al. NBER Working Paper, and the OpenRouter 100T Token Study — are all publicly released research outputs with no individual-level data. Their use requires no ethics approval, as they contain no personal information and are available for academic use without restriction. All benchmark documentation and academic papers used in the inventory are publicly available under standard academic access terms. The project does not involve human participants in any analytical phase, and no personal data is collected, stored, or processed. All processing complies with standard academic research ethics requirements applicable to secondary analysis of publicly available data.

6 Chapter 4: Implementation and Results

6.1 4.1 Phase 1 — Capability Taxonomy Development

6.1.1 4.1.1 Data Collection and Task Instance Extraction

The Phase 1 capability taxonomy was developed through systematic thematic analysis of a corpus of 131 task instances drawn from three primary empirical sources. The sources were selected to provide coverage across commercial enterprise usage, consumer commercial usage, and open-source model usage, ensuring that the resulting taxonomy reflected a broad range of LLM interaction patterns rather than those associated with any single platform or user population.

The primary source was the Anthropic Economic Index (AEI) February 2026 update (Handa et al., 2025), which contributed 103 task instances corresponding to the task categories identified from approximately four million Claude.ai conversations mapped to O*NET Detailed Work Activities. Each AEI task instance was accompanied by a usage percentage indicating the proportion of recorded interactions that involved that task type, making this source the primary anchor for quantitative usage frequency weights in the later gap score analysis. The second source was Ouyang et al. (2025), which contributed 19 task category rows covering writing, coding, analysis, tutoring, and conversational interaction from an analysis of OpenAI platform usage. The third source was the OpenRouter 100T Token Study, which contributed nine category rows from aggregate usage data across a multi-model platform, particularly important for capturing roleplay and open-ended conversational interaction.

6.1.2 4.1.2 Thematic Analysis Outcomes

The thematic analysis produced a taxonomy of eight core capabilities, identified as C01 through C08. The 19 intermediate axial categories produced in the coding process revealed several convergences non-obvious from surface labels. Software engineering, web development, debugging, DevOps, and agentic technical execution all shared the defining characteristic that the expected output was an executable artefact or technical configuration. Professional writing, creative writing, marketing content, and clinical documentation shared the characteristic that the model was the primary author producing a new written artefact. Academic assignment support, tutoring, and concept explanation shared the characteristic that the user's goal was knowledge acquisition rather than task completion. These convergences, governed by the criterion that capabilities should remain distinct only if they require different evaluation approaches, produced the eight-capability structure.

6.1.3 4.1.3 The Eight-Capability Taxonomy

The eight core capabilities and their formal definitions are as follows.

C01 — Content Generation: the capability to produce original, purposeful written or multimedia text artefacts in response to a user's creative, professional, or communicative goal. The model acts as author or co-author, generating content that serves the user's intended audience and context. Sub-categories include academic and educational writing (C01a), professional and business writing (C01b), marketing and promotional writing (C01c), creative writing (C01d), and technical and specialised writing (C01e).

C02 — Code Development and Technical Problem Solving: the capability to write, debug, refactor, and maintain software code; to design and implement technical systems; and to diagnose and resolve technical failures in software, hardware, and networked infrastructure. Sub-categories include web application development (C02a), software engineering and systems (C02b), code debugging and refactoring (C02c), machine learning and AI development (C02d), DevOps and infrastructure (C02e), and technical troubleshooting (C02f).

C03 — Information Retrieval and Advisory: the capability to locate, synthesise, and deliver factual information, evidence-based recommendations, or domain-specific advisory guidance in response to a user's question or decision-making need. Sub-categories include factual and encyclopaedic information (C03a), product and service recommendations (C03b), career, financial, and personal advisory (C03c), research synthesis and literature overview (C03d), and practical how-to guidance (C03e).

C04 — Learning and Education Support: the capability to assist users in acquiring knowledge, skills, or academic qualifications through tutoring, explanation, worked examples, and structured guidance. Sub-categories include academic assignment support (C04a), concept explanation and tutoring (C04b), skill development (C04c), and educational material creation (C04d).

C05 — Review and Feedback: the capability to evaluate, critique, edit, and improve existing written or creative work produced by the user or a third party. The model acts as reviewer, editor, or assessor. Sub-categories include proofreading and grammar correction (C05a), substantive editing and content improvement (C05b), academic feedback and grading (C05c), and peer review and manuscript revision (C05d).

C06 — Translation and Language Processing: the capability to convert text between natural languages, assist users in learning or practising a foreign language, and perform language-specific transformations where the defining feature is a cross-lingual purpose. Sub-categories include document and text translation (C06a), language learning and grammar support (C06b), and multilingual content formatting (C06c).

C07 — Data Analysis and Summarisation: the capability to process, analyse, and synthesise existing datasets, documents, or information corpora to extract insights, patterns, statistical results, or compressed representations. The defining feature is that the user supplies existing data or content as the primary input. Sub-categories include text summarisation and compression (C07a), data analysis and statistical computing (C07b), document processing and format conversion (C07c), and business intelligence and forecasting (C07d).

C08 — Conversational Interaction and Roleplay: the capability to engage in open-ended, interactive dialogue with users for purposes of entertainment, personal support, social practice, or collaborative world-building. The defining feature is that the primary user value lies in the interactive exchange itself rather than in a discrete output artefact. Sub-categories include personal and emotional support dialogue (C08a), interactive roleplay and collaborative fiction (C08b), social and conversational practice (C08c), and entertainment and games (C08d).

The distribution of the 103 Anthropic AEI top tasks across the eight capabilities was as follows: C02 received 35 tasks (34.0%), C03 received 22 tasks (21.4%), C01 received 18 tasks (17.5%), C07 received 11 tasks (10.7%), C04 received 8 tasks (7.8%), C05 received 4 tasks (3.9%), C08 received 3 tasks (2.9%), and C06 received 2 tasks (1.9%).

6.1.4 4.1.4 Validation

The taxonomy was validated against two criteria: coverage of the Anthropic AEI top-task data and inter-coder reliability. All 103 task categories from the AEI were systematically mapped to the taxonomy using the formal decision rules. Of the 103 tasks, 99 were mapped at high confidence on first application of the rules; four tasks were classified at medium confidence because they involved elements of two capabilities and were resolved through the dual-capability disambiguation procedure. No task required the introduction of a new capability category, and no task remained unmapped. Coverage was therefore 103/103 = 100.0%, exceeding the 95% project target.

For inter-coder reliability, a 10% subsample of ten tasks was independently coded by a second coder using only the task description and the formal decision rules. Cohen's was = 1.0000 (P_o = 1.0, P_e = 0.1667), indicating perfect agreement and substantially exceeding the project target of ¿ 0.80 (Cohen, 1960).

6.1.5 4.1.5 Limitations of Phase 1

Three limitations of the Phase 1 analysis are acknowledged. First, the primary coding was conducted by a single researcher, with a second coder involved only in the validation subsample; a complete dual-coding exercise across all 131 task instances would provide stronger inter-rater evidence. Second, the usage frequency weights are derived from a single provider's data and may not fully represent usage patterns on other platforms, particularly developer, enterprise, or non-English-language populations. Third, the AEI data was collected at a specific point in time, and the relative frequencies of different capability types may shift as LLM capabilities and user familiarity evolve.

Capability	Code	Benchmark count
Code Development and Technical Problem Solving	C02	5
Information Retrieval and Advisory	C03	5
Content Generation	C01	4
Data Analysis and Summarisation	C07	4
Learning and Education Support	C04	3
Review and Feedback	C05	3
Conversational Interaction and Roleplay	C08	2
Translation and Language Processing	C06	2

6.2 4.2 Phase 2 — Benchmark Inventory

6.2.1 4.2.1 Search Strategy and Candidate Identification

The Phase 2 benchmark inventory was compiled through a four-channel systematic search designed to identify the major LLM benchmarks in active use as of early 2026. The first channel was the Papers with Code benchmark catalogue, filtered to the Natural Language Processing and Language Models categories. The second channel was the technical reports of major model releases: specifically, the GPT-4 technical report (OpenAI, 2023), the Claude 3 and Claude 3.5 technical reports (Anthropic, 2024), the Gemini technical reports (Google, 2023; 2024), the Llama 2 and Llama 3 technical reports (Meta AI, 2023; 2024), and the DeepSeek R1 technical report (DeepSeek, 2025). Each benchmark cited in two or more of these reports was added to the candidate list. The third channel was a Google Scholar search for "LLM benchmark" and "language model evaluation" filtered to 2020–2025, with inclusion of papers reporting citation counts of 50 or more. The fourth channel was snowball sampling from the reference lists of key project papers.

The four channels collectively identified 127 candidate benchmarks. These were screened against three inclusion criteria: public documentation (a paper or technical report available), use in at least two major model technical reports or equivalent community adoption, and coverage of at least one capability identified in the Phase 1 taxonomy. The screening process selected 28 benchmarks for the final inventory.

6.2.2 4.2.2 Inventory Overview

The 28 benchmarks in the final inventory span all eight capabilities and a publication year range of 2021 to 2025. Table 4.1 presents the distribution by primary capability.

Table 4.1: Benchmark distribution by primary capability

The C02 benchmarks comprise HumanEval/HumanEval+ (Chen et al., 2021; Liu et al., 2023), SWE-bench Verified (Jimenez et al., 2024), LiveCodeBench (Jain et al., 2024), BigCodeBench Hard (Zhuo et al., 2024), and SWE-Lancer Diamond (Miserendino et al., 2025). These range from legacy single-function code generation to frontier agentic software engineering with economic task value, reflecting the evolution of coding evaluation toward realistic software engineering workflows.

The C03 benchmarks comprise MMLU-Pro (Wang et al., 2024), GPQA Diamond (Rein et al., 2023), Humanity's Last Exam (Phan et al., 2025), SimpleQA/FACTS Grounding (Wei et al., 2024; Jacovi et al., 2025), and LiveBench (White et al., 2024), spanning academic knowledge breadth, expert STEM reasoning, frontier expert knowledge, factuality and grounding, and contamination-resistant multi-domain reasoning. The C01 benchmarks comprise IFEval (Zhou et al., 2023), WritingBench (Chen et al., 2025), EQ-Bench Creative Writing v3 (Paech, 2023/2025), and WildBench (Lin et al., 2024). The C04 benchmarks comprise MathDial (Macina et al., 2023), MathTutorBench (Macina et al., 2025), and TutorBench (Srinivasa et al., 2025), each providing tutoring-specific evaluation grounded in pedagogical rubrics.

6.2.3 4.2.3 Quality Ratings Analysis

Each of the 28 benchmarks was rated on five quality dimensions on a 1–5 scale. The mean ratings across all 28 benchmarks were: Coherence 4.61, Relevance 4.54, Accuracy 4.39, Clarity 4.21, and Efficiency 3.50. The Coherence and Relevance scores are highest, reflecting the decision to prioritise benchmarks grounded in real-world workflows. The Efficiency score is lowest, driven by agentic benchmarks (SWE-bench Verified, DA-Code, Spider 2.0, SWE-Lancer Diamond), writing benchmarks requiring LLM judges, tutoring benchmarks requiring human or LLM evaluation, and roleplay benchmarks requiring multi-turn simulation. These benchmarks score low on Efficiency not because they are poorly designed but because their validity depends on realistic evaluation conditions.

6.2.4 4.2.4 Contamination Risk Assessment

Each benchmark was assigned a contamination risk level (High, Medium, or Low). The results were: High risk, 1 benchmark; Medium risk, 3 benchmarks; Low risk, 24 benchmarks. HumanEval (B001) is the single high-risk benchmark, assigned this classification because its 164-problem test set has been publicly available since 2021, is widely distributed in training corpora, and has been noted as saturated in subsequent evaluation literature (Jain et al., 2024). MMLU-Pro, BigCodeBench Hard, and WildBench are the three medium-risk benchmarks. The remaining 24 benchmarks are classified as low risk due to recent release dates, dynamic task refreshing, private holdouts, gated access, or task designs that substantially limit the value of memorised responses.

6.3 4.3 Phase 3 — Coverage Analysis

6.3.1 4.3.1 Coverage Matrix Construction

The Phase 3 coverage matrix was constructed by systematically assessing how well each of the 28 benchmarks tests each of the eight capabilities. The matrix

Rank	Capability	Usage freq.	Coverage score	Gap score	Severity
1	C02 Code Development	0.3019	0.2357	0.2307	High
2	C01 Content Generation	0.2568	0.1429	0.2202	High
3	C03 Info. Retrieval & Advisory	0.1754	0.1714	0.1453	High
4	C04 Learning & Education	0.1113	0.1714	0.0922	High
5	C07 Data Analysis	0.0751	0.2000	0.0601	Medium
6	C05 Review and Feedback	0.0308	0.1286	0.0269	Medium
7	C06 Translation & Language	0.0271	0.0714	0.0252	Medium
8	C08 Conversational/Roleplay	0.0215	0.1286	0.0188	Low

has benchmarks as rows and capabilities as columns; each cell contains a rating from 0 (capability not tested) to 5 (capability tested at excellent quality), with every non-zero cell accompanied by a written justification. Across the full 28 × 8 matrix, 50 cells received non-zero justified ratings.

For each of the eight capabilities, three aggregate coverage metrics were computed: the total coverage score (sum of all cell ratings divided by the maximum possible sum of 140), the benchmark count (number of benchmarks with a non-zero rating), and the average quality (mean rating across non-zero cells).

6.3.2 4.3.2 Gap Score Computation

Usage frequencies for each capability were derived by summing the usage percentages from the Anthropic AEI top-103 task data for all tasks mapped to that capability. C02 had the highest usage frequency at 0.3019, followed by C01 at 0.2568, C03 at 0.1754, C04 at 0.1113, C07 at 0.0751, C05 at 0.0308, C06 at 0.0271, and C08 at 0.0215. The gap scores, computed from these frequencies and the normalised coverage scores, are presented in Table 4.2.

Table 4.2: Gap score ranking — all eight capabilities

A result that merits interpretation is that C02 — Code Development and Technical Problem Solving — ranks first despite having the highest number of benchmarks and the highest normalised coverage score. This is a consequence of the formula structure: C02's gap score is high because its usage frequency (0.3019) is so substantially greater than what even the densest benchmark coverage can proportionally address. The formula rewards benchmarks for their absolute coverage quality but scales the gap by the volume of real-world demand.

C01 ranks second with a gap score of 0.2202. In contrast to C02, C01 has both high demand (0.2568) and relatively thin coverage (0.1429 coverage score, six benchmarks, average quality 3.33). The dominant C01 benchmarks — IFEval and WildBench — measure constraint compliance and real-user task breadth respectively but do not directly evaluate the workplace writing quality, audience adaptation, and professional artefact production that characterise the most common C01 tasks in the AEI data. C03 ranks third with a gap score of 0.1453. The C03 benchmarks are well-designed for testing academic knowledge breadth and expert reasoning but do not test the advisory dimension of C03:

whether models can provide contextually appropriate, evidence-grounded, and appropriately hedged recommendations for users making real decisions.

6.3.3 4.3.3 Statistical Analysis

Three statistical analyses were conducted on the Phase 3 data.

The Pearson correlation between usage frequency and total coverage score across the eight capabilities yielded r = 0.639, p = 0.088 (two-tailed), 95% confidence interval [0.119, 0.927]. The moderate positive correlation indicates a tendency for higher-usage capabilities to receive more benchmark coverage, but the correlation is not statistically significant at the conventional = 0.05 threshold, reflecting substantial uncertainty given the small sample of eight capabilities. The direction of the correlation is consistent with the hypothesis that the research community has directed benchmark development partly toward the capabilities with greatest demand, but the magnitude of the relationship is modest enough that systematic misalignment remains plausible.

The chi-square goodness-of-fit test assessed whether the observed distribution of benchmark counts across capabilities matched the expected distribution if benchmarks were allocated in proportion to usage frequency. The test statistic was ^{2}(7) = 31.96, p ¡ 0.001, Cramér's V = 0.302 (bootstrap 95% CI: [0.202, 0.528]). This result provides strong evidence that the distribution of benchmark coverage across capabilities does not match the distribution of real-world usage demand. C02, which accounts for 30.2% of documented usage, has only 5 primary-capability benchmarks rather than the approximately 8.5 that proportional allocation would predict, while C06, accounting for 2.7% of usage, has 2 benchmarks rather than the 0.76 expected.

The temporal linear regression analyses revealed that C01, C04, and C05 show statistically significant positive slopes, indicating accelerating benchmark activity in these capability areas over the 2020–2025 period. C01 has slope = 0.5143, R^2 = 0.7714, p = 0.0213; C04 has slope = 0.7429, R^2 = 0.8521, p = 0.0086; and C05 has slope = 0.6000, R^2 = 0.7132, p = 0.0344. C02, C03, C07, C06, and C08 show positive but non-significant slopes, indicating increases in benchmark activity that are not distinguishable from noise at the available sample sizes.

6.3.4 4.3.4 Qualitative Deep-Dive Analyses

Four qualitative deep-dive analyses were conducted for capabilities spanning the coverage spectrum.

For C02 (highest gap score, 0.2307), the benchmark landscape is the most active in the inventory, with five primary benchmarks plus secondary coverage from five further benchmarks. Despite this volume, coverage is uneven across the C02 sub-categories. The tasks accounting for the highest usage frequencies in the AEI — troubleshooting hardware and system issues (4.16%), web application development (3.92%), and debugging and refactoring (1.86%) — reflect everyday maintenance and multi-step technical support rather than the competitive programming or high-stakes issue resolution that most existing benchmarks primarily target. The aggregate quality score of 3.30 for C02 benchmarks reflects this unevenness.

For C07 (medium gap, 0.0601), the benchmark landscape includes four primary benchmarks with the highest average quality score in the inventory (4.00) and reasonable coverage of data analysis workflows. The coverage limitation is structural: existing C07 benchmarks tend to be CSV-centric, SQL and business intelligence-centric, or document question-answering-centric. Real data analysis tasks often involve messy spreadsheets, inconsistent labels, missing values, mixed file formats, and the need to communicate cautious conclusions to non-technical audiences — a full analyst workflow not cleanly captured by any single benchmark.

For C04 (high gap, 0.0922), the evaluation landscape has grown substantially since 2023, driven by the introduction of MathDial, MathTutorBench, and TutorBench. MathTutorBench in particular demonstrates that stronger academic reasoning models are not necessarily better tutors, providing evidence that the tutoring construct is distinct from general knowledge or reasoning performance. The temporal regression result for C04 (slope = 0.7429, p = 0.0086) indicates that this is the area of most rapid benchmark development in the recent period.

For C06 (medium gap, 0.0252), the benchmark landscape is served by only two primary benchmarks — BenchMAX and WMT24++ — but both are of high quality (average rating 5.00). The relatively low gap score reflects both modest usage frequency in the AEI data and the maturity of machine translation evaluation as a research area. The main uncovered areas within C06 are language learning support and multilingual instruction following.

6.3.5 4.3.5 Case Studies — Benchmark Performance Versus Real-World Failure

Six case studies were documented to illustrate instances where high benchmark performance did not correspond to reliable performance in realistic deployment settings.

Xu et al. (2025) demonstrate that frontier LLM agents complete approximately 30% of integrated workplace tasks in the TheAgentCompany benchmark despite strong standard benchmark performance, with failure modes including long-horizon planning failures and tool use errors. Jain et al. (2024) demonstrate through LiveCodeBench that model performance on coding problems declines on problems released after the model's likely training cutoff, providing direct evidence of contamination effects in static coding benchmarks. Pezeshkpour and Hruschka (2024) show that changing the order of answer options in MMLU-style questions shifts model rankings by up to eight positions, demonstrating that leaderboard rankings based on multiple-choice benchmarks are partly a function of presentation format.

Kanjee et al. (2024) document that LLMs achieving strong results on medical knowledge benchmarks continue to make clinically significant errors in realistic patient case scenarios, demonstrating the gap between aggregate knowledge performance and capability for consequential real-world decision support. Magesh et al. (2025) document that legal research tools based on general LLMs generate fictitious legal citations that appear authoritative, illustrating why factuality benchmarks measuring short-fact recall are insufficient for advisory tasks carrying the greatest real-world consequence. Singh et al. (2025) demonstrate that systematic selective disclosure of benchmark results distorts leaderboard rankings by up to 112 positions, representing a structural validity problem in the benchmark reporting ecosystem.

The cross-case pattern is consistent: benchmarks are valid instruments for measuring specific capabilities under specific conditions, but they are regularly used as broader evidence of capability than their design and context can support.

6.3.6 4.3.6 Limitations of Phase 3

Four limitations of the Phase 3 analysis are acknowledged. First, all coverage ratings in the matrix are researcher judgements rather than experimentally verified measures; different researchers applying the same rubric might assign different ratings to some cells. Second, the usage frequency weights are derived from a single provider's data, which may not represent the full distribution of LLM use across different providers and user populations. Third, the Pearson correlation and chi-square tests are based on eight capability-level observations, a small sample for statistical inference; the results should be treated as descriptive summaries rather than confirmatory tests of population-level hypotheses. Fourth, the temporal regressions use only six annual observations, making individual year counts highly influential and slope estimates correspondingly uncertain.

7 Chapter 6: Recommendations and Toolkit

7.1 6.1 Overview

This chapter presents the Phase 5 outputs: five benchmark design specifications targeting the highest-priority coverage gaps identified in Chapter 4, improvement recommendations for the 28 existing benchmarks in the Phase 2 inventory, the practitioner Assessment Toolkit, and an implementation roadmap for the research community. The benchmark specifications and improvement recommendations respond directly to the Phase 3 gap score analysis, which identified Code Development and Technical Problem Solving (C02, 0.2307), Content Generation (C01, 0.2202), Information Retrieval and Advisory (C03, 0.1453), Learning and Education Support (C04, 0.0922), and Data Analysis and Summarisation (C07, 0.0601) as the five highest-priority gaps.

7.2 6.2 Benchmark Design Specifications

7.2.1 6.2.1 Selection Rationale

Five benchmark specifications were developed, one for each of the five highest-priority gaps. The selection criterion for specifying a new benchmark rather than only recommending improvements to existing ones was whether the gap score analysis, combined with the qualitative deep-dive assessment, indicated that existing benchmarks were structurally unable to address the evaluation need — either because their task design was too narrow, their scoring mechanism was misaligned with the capability construct, or the capability area had no substantive benchmark coverage addressing its most common use cases. All five selected capabilities met this criterion. No two proposed benchmarks target the same capability, ensuring that the specifications collectively expand coverage across the identified priority gaps.

7.2.2 6.2.2 MaintBench — Software Maintenance Benchmark (C02)

MaintBench is proposed for C02, Code Development and Technical Problem Solving, which has the highest gap score in the project (0.2307). The benchmark is designed to test whether models can diagnose, modify, integrate, and maintain realistic software systems rather than only solving isolated coding prompts. The focus is on multi-file debugging, dependency integration, refactoring under constraints, configuration repair, regression prevention, and handoff-quality explanations — the middle layer of everyday software maintenance that is underrepresented in existing C02 benchmarks.

The existing C02 landscape provides strong coverage of isolated code generation, competitive programming, library-oriented code tasks, and agentic repository-level issue resolution. What remains uncovered is the practical daily maintenance work that accounts for the highest-frequency AEI task categories: troubleshooting hardware and system issues (4.16% of usage), developing and debugging web applications (3.92%), and debugging and refactoring code across languages (1.86%). Each MaintBench task provides the model with a repository snapshot, a stakeholder request, relevant logs or failing tests, and project constraints. The required output is a patch accompanied by an engineering note explaining the diagnosis, the change, the verification steps, and any remaining risks.

The proposed first release contains approximately 600 tasks across 60 repositories, with each repository contributing 8–12 tasks. Evaluation combines automated correctness (tests, static analysis, build execution) with expert-judged assessment of maintainability, diagnosis quality, and constraint adherence. Implementation requires 4–6 months, engineering research expertise, containerised execution infrastructure, and expert developer time for hidden test construction and maintainability review.

7.2.3 6.2.3 WorkWriteBench — Workplace Writing Benchmark (C01)

WorkWriteBench is proposed for C01, Content Generation, which has the second-highest gap score (0.2202). The benchmark is designed to evaluate professional writing under realistic workplace constraints, including professional communications, decision documents, marketing copy, technical documentation, and audience-specific transformations. None of the existing C01 benchmarks addresses the specific evaluation challenge of workplace professional writing: the requirement that outputs preserve source facts accurately, fit a specified audience, adhere to tone and format constraints, and avoid unsupported claims.

Each WorkWriteBench task provides the model with a writing brief, source facts, a target audience, a purpose, format requirements, and constraints. The proposed first release contains approximately 1,000 tasks across six task families. Evaluation uses hybrid scoring combining automated checks (required facts present, prohibited claims absent, length constraints, format requirements) with human or validated LLM-judge assessment of audience fit, tone, and writing usefulness. Implementation requires 3–4 months at medium cost, with human annotator time as the primary cost driver.

7.2.4 6.2.4 GroundedAdviceBench — Advisory Reasoning Benchmark (C03)

GroundedAdviceBench is proposed for C03, Information Retrieval and Advisory, which has the third-highest gap score (0.1453). The benchmark is designed to evaluate whether models can provide factual, evidence-grounded, context-sensitive advice while communicating uncertainty and maintaining appropriate limits — the advisory dimension of C03 that existing knowledge benchmarks do not directly measure. The advisory construct is central to the AEI task categories mapping to C03, including medical information (2.38% of usage), career advisory (1.63%), and product comparison (2.25%).

Each GroundedAdviceBench task provides the model with a user question, user context, an evidence packet, and response requirements. Evidence packets may include product tables, policy documents, medical or legal guidance excerpts, research abstracts, or deliberately conflicting or incomplete information. The proposed first release contains approximately 800 tasks across five advisory domains. Evaluation uses rubric scoring across factuality, grounding, uncertainty communication, relevance, and safety dimensions. The benchmark distinguishes itself from MMLU-style knowledge benchmarks by making advisory behaviour the primary construct rather than a by-product of knowledge recall.

7.2.5 6.2.5 TutorScaffoldBench — Tutoring and Pedagogy Benchmark (C04)

TutorScaffoldBench is proposed for C04, Learning and Education Support, which has the fourth-highest gap score (0.0922). The benchmark is designed to evaluate models as tutors rather than as answer engines, assessing misconception diagnosis, adaptive scaffolding, formative feedback, concept explanation, and academic-integrity-preserving responses. TutorScaffoldBench does not replace MathDial, MathTutorBench, or TutorBench but extends them by increasing subject diversity, making pedagogical sub-skill reporting more interpretable, and explicitly covering academic-integrity scenarios where the appropriate tutor response is to refuse answer delivery and offer learning-support alternatives.

The proposed first release contains approximately 1,200 tasks across seven subject areas, with mathematics representing at most 30% of the dataset to avoid over-specialisation. Evaluation uses pedagogical rubric scoring across misconception diagnosis, scaffolding quality, formative feedback, and adaptive follow-up dimensions, with educator review for calibration. Implementation requires 3–4 months at medium cost.

7.2.6 6.2.6 MessyDataBench — Data Analysis Benchmark (C07)

MessyDataBench is proposed for C07, Data Analysis and Summarisation, which has a gap score of 0.0601 and Medium severity. Despite relatively stronger coverage than the top four capabilities, existing C07 benchmarks do not collectively represent the full analyst workflow. They tend to be CSV-centric, SQL-centric, or document QA-centric and do not systematically test the model's ability to handle imperfect data, communicate cautious conclusions, or support non-technical stakeholders.

Each MessyDataBench task provides the model with a user request, one or more data files or document excerpts containing realistic data-quality issues, and a communication target. The benchmark focuses on the full analyst workflow: understanding the data, handling quality issues, applying appropriate methods, interpreting uncertainty, and communicating responsibly. The proposed first release contains approximately 700 tasks with associated data packets. Evaluation combines automated calculation correctness checks with rubric scoring for method fit, uncertainty communication, and stakeholder-appropriate reporting.

7.3 6.3 Improvements to Existing Benchmarks

In addition to the five new benchmark proposals, actionable improvements were identified for each of the 28 benchmarks in the Phase 2 inventory. For the C02 benchmarks, HumanEval should be retained only as a historical anchor with explicit high-contamination-risk labelling in any comparative reporting; SWE-bench Verified should add sub-category reporting by bug type, repository complexity, and maintenance burden; LiveCodeBench should improve clarity about the distinction between competitive programming skill and general software engineering; BigCodeBench Hard should expand language and ecosystem coverage beyond Python; and SWE-Lancer Diamond should publish fuller task metadata and evaluation harness documentation for reproducibility.

For the C03 benchmarks, MMLU-Pro should add robustness reporting for answer-order sensitivity; GPQA Diamond should publish domain-specific uncertainty reporting to avoid overgeneralisation to advisory tasks; SimpleQA and

FACTS Grounding should extend toward evidence-grounded recommendations and uncertainty handling; and LiveBench should improve capability attribution to allow disaggregation of multi-domain scores by capability. For the C01 benchmarks, IFEval should add longer, source-grounded writing tasks; WritingBench should strengthen judge calibration documentation; and WildBench should add capability-specific subset reporting. For the C04, C05, C06, and C08 benchmarks, the primary recommendations are to improve reporting granularity, validate LLM judges against human ratings where they are used as the primary scoring mechanism, and publish annotator guidelines and inter-rater reliability statistics to enable reproducibility.

7.4 6.4 The Assessment Toolkit

7.4.1 6.4.1 Design Rationale

The Assessment Toolkit addresses a practical problem that the benchmark specifications and gap scores cannot resolve independently: practitioners selecting benchmarks for a specific deployment context need a way to evaluate coverage for their own use-case portfolio, not for the aggregate distribution of all LLM use. A hospital AI deployment has a different capability priority profile from a software development assistant or a consumer information service, and the population-average gap scores in Chapter 4 may not correspond to any specific deployment context. The toolkit operationalises the Phase 1 taxonomy and Phase 3 coverage data as a structured instrument that practitioners can use interactively. Its intended users are model evaluators, AI deployment leads, and researchers who need to assess whether a proposed benchmark suite covers the capabilities relevant to their context.

7.4.2 6.4.2 Toolkit Components

The Assessment Toolkit is implemented as a seven-tab Excel workbook accompanied by a PDF documentation guide. The Instructions tab provides a step-by-step guide to using the toolkit, including prerequisites, a glossary, and a summary of how to interpret output scores. The Capability_Checklist tab presents the eight capabilities with brief plain-language descriptions and subcategory breakdowns; users select dropdown menus to indicate which capabilities are relevant to their deployment context and assign priority weights. The Benchmark_Profiles tab contains one row per benchmark from the Phase 2 inventory with key fields from the benchmark database. The Quality_Rubric tab presents the five-dimension quality rubric with definitions and scoring guidance and can be used to assess benchmarks not in the Phase 2 inventory, extending the toolkit's usefulness as the benchmark landscape evolves.

The Coverage_Calculator tab is the computational core: users input their capability priorities, and the calculator automatically computes a weighted coverage score for any selection of benchmarks from the Phase 2 inventory. The formula sums the coverage matrix ratings for the selected benchmarks across the user's priority capabilities, weights them by the user's stated priority levels, and expresses the result as a proportion of the maximum possible weighted coverage. The Worked_Examples tab presents three worked case studies demonstrating the toolkit in action: a medical information service use case, a software development assistant use case, and a consumer writing assistant use case. The Interpretation_Guide tab provides guidance on reading the coverage calculator outputs, including limitations and instructions for updating the toolkit as new benchmarks are published.

7.4.3 6.4.3 Worked Example

To illustrate the toolkit in use, consider a team deploying an LLM to provide general health information to the public. They identify their primary capability priorities as C03 Information Retrieval and Advisory (High priority), C04 Learning and Education Support (Medium priority, for patient education materials), and C05 Review and Feedback (Low priority, for reviewing draft patient communications). C01, C02, C06, C07, and C08 are marked Not Applicable. The Coverage_Calculator computes the weighted coverage score for a proposed benchmark suite comprising MMLU-Pro, GPQA Diamond, SimpleQA/FACTS Grounding, and LiveBench. The output shows strong coverage of C03a (factual knowledge) but no coverage of the advisory dimension of C03 (contextual recommendation, uncertainty communication, safety boundary-setting) or of C04 tutoring capability. The weighted coverage score for the proposed suite is 0.44 against the user's capability profile, below the recommended threshold of 0.60, and the toolkit recommends supplementing with a C03 advisory-focused benchmark or acknowledging the advisory gap explicitly in evaluation reporting.

7.5 6.5 Implementation Roadmap

The implementation roadmap translates the Phase 5 outputs into a realistic timeline for the research community, structured across three horizons. In the short term (0–1 year), the immediate priority is to convert the five benchmark specifications into pilot-ready artefacts. MaintBench and WorkWriteBench should be first, as they address the two largest usage-weighted gaps and have relatively tractable development timelines of 3–6 months each. Concurrently, existing benchmarks should adopt the low-cost improvements identified in Section 6.3: contamination risk labelling, sub-category reporting, and judge validation documentation.

In the medium term (1–3 years), the strongest pilots should be converted into maintained benchmarks with public development splits, private or refreshed evaluation splits, and documented inter-rater reliability. The research community should move toward requiring at least one benchmark per high-usage capability in frontier model technical reports, ensuring that C01, C03, and C04 — currently underrepresented in major model release evaluations — receive coverage equivalent to C02 and C03 knowledge benchmarks.

In the long term (3+ years), the goal is a capability-weighted benchmark ecosystem in which evaluation reports communicate model fitness for specific capability profiles rather than aggregate leaderboard positions. This requires living benchmark suites with private holdouts and regular task refreshes, community standards for LLM judge validation, and benchmark reporting standards that separate capability-level performance from aggregate scores. The Assessment Toolkit provides a partial model for what this would look like in practice.

7.6 6.6 Limitations

Three limitations of the Phase 5 outputs are acknowledged. First, the benchmark specifications are design documents, not implemented benchmarks. Their validity as evaluation instruments cannot be confirmed until they are built, piloted, and subjected to empirical validation. Design decisions made during implementation — task distribution, scoring rubric calibration, choice of representative repositories or domains — will require iteration beyond what can be anticipated at the design stage. Second, the Assessment Toolkit is based on the Phase 2 benchmark inventory and Phase 3 coverage matrix as of early 2026. As new benchmarks are published, the toolkit will require updating; without maintenance, coverage scores will become stale. Third, the implementation roadmap is indicative rather than binding. The timelines are based on the researcher's assessment of development feasibility given the task types, dataset sizes, and evaluation methodologies specified, but actual implementation depends on the availability of research funding, institutional support, and community adoption.

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